

Interpretable Difficulty Estimation for Classical Guitar from Symbolic Notation

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1 Introduction and Motivation

Selecting repertoire that matches a student’s current technical level is a central task in music learning. In classical guitar, difficulty judgments influence lesson planning, sequencing of repertoire, practice efficiency, and student motivation, yet these judgments are often based on subjective teacher experience and informal consensus rather than transparent and reproducible criteria. Graded repertoire systems such as GuitarBurst[1] show that there is clear practical demand for structured difficulty information in the classical guitar domain, but they do not automatically explain how score level technical characteristics translate into a given difficulty level.

Recent work in music information retrieval has shown that symbolic music representations such as MusicXML can support automatic difficulty estimation. Yaroch used machine learning models to estimate score difficulty from symbolic features extracted from MusicXML[5], and Ramoneda et al. showed that difficulty estimation can also be approached through a small set of interpretable descriptors organised in a rubric like model [2]. However, most of this work has focused on piano repertoire, and the currently strongest interpretable approach, RubricNet [2], uses descriptors tailored to keyboard performance rather than guitar technique.

This creates a clear research gap. Guitar specific work has begun to formalise playability, particularly for rhythm guitar chord progressions, through descriptors such as barre usage, chord fingering difficulty, chord transitions, and right-hand complexity[4]. Yet there is still no widely used interpretable difficulty model aimed at classical guitar solo repertoire that combines symbolic analysis with guitar specific technical descriptors and an external graded repertoire resource such as GuitarBurst[1]. The motivation of this project is therefore to design a model that does not only predict a difficulty level, but also provides a structured explanation of why a classical guitar piece is difficult. The project aims to adapt recent explainable difficulty-estimation methods developed for piano to the domain of classical guitar, grounding the descriptor design in the actual technical demands of the instrument and in established graded repertoire resources.

2 Research Questions and Research Goals

The project is guided by the following research questions:

1. Which guitar-specific symbolic descriptors best capture the technical difficulty of classical guitar pieces as reflected in graded repertoire resources such as GuitarBurst[1]?
2. Can an interpretable rubric based model using these descriptors predict piece level classical guitar difficulty with useful accuracy?
3. Does combining multiple dimensions of difficulty, such as left hand technique, right hand technique, and global score complexity, improve prediction compared with any single dimension alone?
4. To what extent do the model’s descriptor contributions align with common pedagogical intuitions about what makes a classical guitar piece difficult?

The research goals follow directly from these questions.

1. The project aims to define a compact set of guitar specific, interpretable descriptors that can be computed from symbolic notation.
2. It aims to build a piece level difficulty classifier for classical guitar based on these descriptors.
3. It aims to evaluate whether a multidimensional feature design, combining separate descriptors for left hand technique, right hand technique, and global score complexity performs better than one dimensional alternatives.
4. It aims to show that interpretability is not only technically possible but pedagogically meaningful in the context of classical guitar education.

Taken together, these goals define the contribution of the work as a compact, interpretable difficulty-estimation framework for classical guitar that combines symbolic score analysis, guitar-specific descriptors, and a rubric-like model architecture.

3 Related Work

A useful starting point is Yaroch’s thesis[5]. Yaroch used MusicXML based feature extraction together with linear regression and convolutional neural networks to estimate difficulty measure by measure, showing that symbolic notation can support automatic difficulty analysis. That work is important because it confirms the feasibility of symbolic score difficulty prediction, but it remains instrument agnostic, is trained on a small dataset, and relies on self-assigned difficulty ratings rather than a widely accepted grading source.

A more relevant methodological foundation comes from Ramoneda et al. [3], the authors argue that difficulty is multidimensional and depends on notation,

technique, and expressiveness rather than on a single type of feature. They show that combining different dimensions improves difficulty prediction for piano repertoire, which strongly supports a guitar model that separates left hand demands, right hand demands, and global score complexity.

The most relevant work is Ramoneda et al.’s[2]. This paper introduces RubricNet, a white-box model in which each descriptor contributes transparently to the final difficulty estimate, much like a teaching rubric. RubricNet is attractive for the present project because it offers a realistic and explainable alternative to black box neural models. At the same time, its descriptors are largely piano oriented, which opens the possibility of adapting the framework to guitar specific technique.

On the guitar side, Vasquez et al.[4] provide an important precedent for descriptor design. That paper develops a rubric based playability metric for rhythm guitar using expert informed criteria such as uncommonness of chords, chord finger positioning, barre difficulty, right hand complexity, and rhythmic regularity. Although its repertoire domain is pop song chord playing rather than classical solo guitar, it demonstrates that guitar difficulty can be formalised in interpretable terms. This supports the idea that a classical guitar difficulty model should use descriptors grounded in actual guitar technique rather than generic symbolic complexity alone.

Taken together, these studies suggest a promising synthesis: symbolic score analysis from Yaroeh, a multidimensional view of difficulty from Ramoneda’s CIPI work, an explainable descriptor-based architecture from RubricNet, and guitar-specific rubric design from the chord-playability literature.

4 Method

The proposed method is to build an interpretable, rubric based piece level classifier for classical guitar difficulty. The input will be symbolic representations of classical guitar pieces, ideally in MusicXML form, and the output will be a predicted difficulty class derived from an external graded repertoire source such as GuitarBurst[1].

4.1 Dataset and Labels

The dataset will consist of a curated set of classical guitar pieces with published difficulty labels. GuitarBurst[1] is the preferred primary source for the pieces to be used and their difficulty classification because it provides a large graded repertoire database for classical guitar and is explicitly intended to support repertoire selection. In addition, selections from existing symbolic guitar datasets such as GAPS (Guitar Aligned Performance Scores) and the DADA-GP GuitarPro collection will be used as score sources, with GuitarBurst difficulty levels providing the target labels wherever pieces can be matched across datasets. Since GuitarBurst[1] uses a fine grained 1–20 difficulty scale, these labels may be grouped into fewer ordinal classes to reduce sparsity and make evaluation more robust.

4.2 Descriptor Design

The project will define a compact descriptor set organised into three dimensions:

- Left-hand technique, for example barre usage, maximum and average stretch span, position-shift frequency, and chord grip complexity.
- Right-hand technique, for example arpeggio-pattern variation, string-crossing irregularity, and the presence of special techniques where available in the notation.
- Global score complexity, for example note density, polyphony, rhythmic complexity, and other symbolic descriptors adapted from previous difficulty estimation work.

The emphasis will be on descriptors that are small in number, computable from notation, and understandable to teachers and students. In this sense, the descriptor set is intended to function like a guitar pedagogy rubric rather than a large opaque feature vector.

4.3 Model

The main model will follow the logic of RubricNet: a small feed-forward neural network in which each descriptor is passed through its own linear layer with a non-linear activation (e.g. $\tanh$), and the resulting contributions are summed into an aggregate difficulty score that is then mapped to an ordinal difficulty class. This architecture is parameter-efficient and makes it possible to inspect the contribution of each descriptor for any individual piece, similar to a rubric in music pedagogy.

To ensure a fair comparison between models, hyperparameters will be tuned systematically, but within a scope appropriate for a bachelor thesis. For the RubricNet-style neural model, key hyperparameters such as learning rate, number of hidden units, dropout rate, and batch size will be selected via a small grid or random search over a set of reasonable candidates, using cross-validation on the training data. For simpler baseline models (e.g. ordinal regression or tree-based classifiers), central hyperparameters such as regularisation strength, maximum tree depth, or number of trees (in the case of a Random Forest) will be tuned in an analogous way.

As a first baseline, an ordinal regression model will be implemented on the same descriptor set, providing a simple, linear and transparent comparison model. In addition, a tree-based method such as a decision tree or Random Forest classifier will be used as a second baseline, since such models often achieve robust performance and offer model-level interpretability through feature importance measures.

If time permits, a small fuzzy-logic classifier will also be explored as an alternative, highly interpretable approach. In this case, difficulty would be modeled through fuzzy rules formulated in natural language (e.g. “if barre frequency is high and position shifts are large, then difficulty is rather high”).

This would allow a direct comparison between a data-driven rubric model and an explicit rule-based system.

5 Evaluation Plan

The evaluation will address both predictive performance and interpretability.

5.1 Predictive Evaluation

The main predictive evaluation will compare several model variants:

- A model using only left hand descriptors.
- A model using only right hand descriptors.
- A model using only global score descriptors.
- A model using left hand + right hand descriptors.
- A full model using all three dimensions together.

This comparison directly addresses whether combining multiple dimensions of difficulty improves prediction over any single dimension alone, following the logic suggested by Ramoneda et al.’s multi dimensional piano work[3]. Performance will be evaluated using ordinal classification metrics such as accuracy, mean absolute error in predicted class, and confusion patterns between neighboring difficulty levels.

5.2 Interpretability Analysis

Beyond prediction accuracy, the project will analyse which descriptors contribute most strongly to the predicted difficulty of each piece and how those contributions vary across difficulty levels. If possible, this analysis will be supplemented by a qualitative check against pedagogical expectations: for instance, whether pieces at higher levels indeed receive stronger contributions from barres, position shifts, polyphony, or right-hand complexity.

In addition, the importance of entire feature groups (left-hand, right-hand, global score complexity) will be investigated using a group level feature selection analysis inspired by multi objective approaches. The idea is to search for small feature subsets that maintain high classification performance while minimising the number of descriptors from a given group; if performance drops markedly when a group is constrained, that group is considered important, whereas similar performance with a strongly reduced group indicates lower importance.

This part of the evaluation is important because the goal of the project is not only to predict difficulty, but also to make the model’s reasoning meaningful to teachers and learners. In this way, the evaluation will address both methodological questions (can explainable descriptor-based models transfer from piano to guitar?) and pedagogical questions (does the model provide insights that are useful for repertoire selection and teaching practice?).

6 Timeline

May-June 2026 Finalise literature review and descriptor list; select and pre-process datasets, draft initial versions of Introduction and Related Work.

June-July 2026 Implement feature extraction pipeline and initial baselines.

July-August 2026 Implement RubricNet-style model and run first experiments.

August-September 2026 Hyperparameter tuning, ablation studies (single vs multi-dimension), start writing Methods and Evaluation sections.

September 2026 Refine all chapters, and prepare final presentation.

References

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